

MATTHEW SNOBY

Cloud Infrastructure & Platform Engineering Technical Leader

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PROFESSIONAL SUMMARY

Technical leader with 25+ years spanning cloud platforms, secure developer infrastructure, Kubernetes, CI/CD, observability, AI-assisted operations, and embedded Linux. Builds durable systems, modernizes critical platforms without service disruption, and remains hands-on from architecture through production support.

SELECTED IMPACT

AI & Knowledge Systems	Built Codex-assisted FedRAMP workflows and on-call triage; designed and operate DIP, a FastAPI document ingestion and hybrid retrieval platform using Qdrant, SQLite FTS5, reranking, and cited answers.
Reliable Developer Platforms	Designed and operated Drone CI/CD, Terraform Enterprise, Vault-backed secrets, hardened build images, and GitOps workflows supporting mission-critical Webex engineering.
Production Operations	Owned a Splunk platform serving about 300 Webex users and run mining infrastructure with Prometheus, Grafana, Loki, PagerDuty, and PostgreSQL 18 Patroni high availability.

PROFESSIONAL EXPERIENCE

Cisco / Webex | Cloud Engineering Technical Leader | Sep 2010 - Present

Technical leadership across AI-assisted infrastructure, cloud platforms, secure build systems, observability, DRM, and embedded architecture.

- Designed, built, deployed, and operate DIP, converting mixed office documents into searchable engineering knowledge with parallel conversion, SHA256 deduplication, sentence-aware chunking, batched embeddings, Qdrant vectors, SQLite FTS5, hybrid retrieval, reranking, and cited answer generation.
- Built Codex automations for a FedRAMP-secure Kubernetes image update workflow, reducing a 20+ step GitOps/JIRA/MFA-heavy process to enabling the automation and confirming MFA; also applied Codex to runbook-guided on-call investigation.
- Designed and deployed Drone CI/CD and Terraform Enterprise for Webex Logging Metrics, integrated build and deployment secrets with HashiCorp Vault, and maintained the architecture in production for more than five years without major redesign.
- Designed Vault hierarchies, roles, and hand-written policies for CI, development, staging, and production; advised a team implementing GitHub Actions on secrets, images, reliability, deployment flow, and day-two operations.
- Owned a production Splunk platform for approximately 300 Webex users and drove Ubuntu 20.04-to-24.04 hardening, DUO SSH, secure build evidence, and image migrations across AWS and OpenStack/3AZ.
- Moved internal workloads to Kubernetes on AWS before EKS, automating multi-region clusters with Ansible and KOPS; designed a dedicated business-critical platform for Webex bot workloads with Kube2IAM and Vault isolation.
- Earlier Cisco leadership included RDK and Android set-top box architecture, GStreamer DRM pipeline plugins, Intertrust DRM and Nagravision integrations, and open source collaboration with Linaro LHG.

Vtlt | Principal Engineer | Sep 2009 - Sep 2010

Cisco third-wave DRM expert and Linux distribution architect.

Scientific Atlanta / Cisco Systems | Associate Staff Engineer | Aug 2003 - Sep 2009

Led low-level platform work across dual-processor set-top box systems, Linux kernel drivers, DOCSIS, DMA, co-processors, and hardware bring-up.

- Owned board support package bring-up for dual-processor SPARC ASIC platforms, led driver design and code reviews, and carried platform delivery responsibility for Cablevision deployments.
- Designed multi-CPU DMA functionality, Linux character/network/proprietary drivers, DOCSIS co-processor boot flows, and DOCSIS-over-USB and 802.11b proof-of-concept systems.

EARLIER EXPERIENCE

Livewire | Digital Television Electrical Engineer | Jan 2001 - Aug 2003

Designed embedded set-top box drivers and resident applications; integrated Nagravision conditional access, parsed MPEG/DVB service information, and worked across TCP/IP, Win32, MIPS, ARM, DSP, and ST5518 platforms.

Barco | Software Engineer | May 2000 - 2001

Developed 8051 embedded C software for broadcast systems that digitized and transported audio/video over fiber, including requirements and LCD interface redesign.

INDEPENDENT ENGINEERING

Vipor Mining Pool | Infrastructure, Observability & Database Reliability | Independent Project

- Built and operate production mining-pool infrastructure spanning blockchain nodes, stratum services, HAProxy/Nginx routing, Cloudflare/DNS, centralized Vector.dev-to-Loki logging, Prometheus/Grafana monitoring, Alertmanager, and PagerDuty.
- Operate PostgreSQL 18 with Patroni primary/replica high availability; migrated from RDS to EC2 and then colocation without downtime using logical replication while supporting millions of dollars in pool revenue.

Cuckoo Mining Algorithm | World-Fastest CPU Miner | Infrastructure R&D

- Wrote the world's fastest CPU miner for the Cuckoo mining algorithm by combining affinity and NUMA optimization, compiler tuning, low-level profiling, and benchmark-driven algorithm improvements on Ryzen and EPYC systems.
[LinkedIn article: Cuckoo CPU miner performance work](#)

Local AI Systems Lab | Private LLM & Retrieval Infrastructure | AI Infrastructure R&D

- Run Qwen 3.x models through llama.cpp with GGUF, MTP speculative decoding, long-context and KV-cache tuning; evaluate private coding and operations workflows across multi-GPU and single-GPU systems.

TECHNICAL SKILLS

Cloud & Platforms: AWS, OpenStack, Kubernetes/KOPS, Terraform, Ansible, Vault, hardened Linux images

Developer Infrastructure: Drone CI, GitHub Actions, Jenkins, CircleCI, GitOps, secure build systems

Observability & Data: Splunk, Prometheus, Grafana, Loki, Vector.dev, PagerDuty, PostgreSQL, Patroni, Qdrant, SQLite FTS5

Systems & AI: C/C++, Python, Linux kernel drivers, DOCSIS, DRM/conditional access, FastAPI, local LLM inference, GGUF, RAG, reranking

EDUCATION & SELECTED LINKS

B.S. Computer Engineering, Southern Polytechnic State University, Marietta, Georgia | 2000

[Vipor.net](#) | [Chromium/CDM Presentation](#)